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ABSTRACT

The combined paleomagnetic results obtained from all formations of the Karroo Supergroup of Madagascar are compared with corresponding data obtained in Africa and South America. This provides unequivocal evidence in favor of the reconstructions of du Toit and Smith and Hallam.

INTRODUCTION

Studies of marine magnetic anomalies in the Indian Ocean (McKenzie and Sclater, 1971; Sclater and Fisher, 1974; Schlich and Fondeur, 1974; Schlich, 1975) have mapped the age of the ocean floor back to Late Cretaceous time, about 75 m.y. ago. These studies have enabled workers to establish the history of the formation of the present Indian Ocean relating mostly to the northward drift of India (McKenzie and Sclater, 1971; Sclater and Fisher, 1974). Unfortunately, the data do not go far enough back in time to make it possible unambiguously to reconstruct the continents around the Indian Ocean as they were in the ancient supercontinent of Gondwanaland. Crucial in deciding upon this reconstruction is the paleoposition of Madagascar. The exact relationship of the eastern parts of Gondwanaland (Antarctica, Australia, and India) to the western parts (Africa and South America) depends very much on where Madagascar is reconstructed relative to the opposing African continental margin.

Three possibilities have long been considered for the paleoposition of Madagascar: (1) a position adjacent to east Africa off the coast of modern Somalia, Kenya, and Tanzania as originally proposed by du Toit (1937) and favored by Smith and Hallam (1970) in their geometrical reconstruction (rotation of Madagascar about an Euler pole at 9°S, 313°E by an angle of 15°); (2) a position adjacent to Mozambique in the coastal embayment in southeast Africa, originally proposed by Wegener (1929) and supported with geological arguments by Flores (1970); Flores' reconstruction is found by rotating Madagascar about an Euler pole at 48°S, 70°E by an angle of 14° (Embleton and McElhinny, 1975); and (3) the same position as at present.

Deep-sea drilling in the Indian Ocean around Madagascar on Leg 25 of the Deep Sea Drilling Project (Schlich and others, 1975) has not clarified the problem much further except to indicate that Madagascar has not occupied the northerly position off east Africa for the past 90 m.y. In an attempt to resolve this question, we have carried out paleomagnetic measurements on sedimentary rocks of the Karroo Supergroup of Madagascar (Embleton and McElhinny, 1975; Razafindrazaka and others, 1976; McElhinny and Embleton, 1976). Although these studies were carried out independently at the two laboratories at Canberra and at St. Maur, we find that there has been no duplication of results from the different formations; our combined results cover virtually the entire time span represented by the formations of the Karroo Supergroup.

PALEOMAGNETIC DATA

The sequence of the Karroo Supergroup of Madagascar ranges from late Carboniferous to Middle Jurassic in age and has been divided by Besairie (1972, 1973) into three groups separated by unconformities: (1) Isalo Group—Middle Triassic to Middle Jurassic, (2) Sakamena Group—Upper Permian to Lower Triassic, and (3) Sakoa Group—upper Carboniferous to Lower Permian. The Sakoa Group, locally developed in the southwest of the island, commences with the Glacial Series (Sakoa I), containing tillites interbedded with green and black shales about 400 m thick. The Coal Measures (II) follow, overlain by the Lower Red Series (Sakoa

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TABLE 1. SUMMARY OF PALEOMAGNETIC DATA FROM KARROO SUPERGROUP OF MADAGASCAR

Group	Age	Locality (°S)	Locality (°E)	Site	No. of samples	Dec. (°)	Inc. (°)	Pole position (°N)	Pole position (°E)	k	α_{95}	Ref.
Isalo III	Middle Jurassic	16.4	47.2	2	8	351	-43.5	77	268			1
Isalo II	Early Jurassic	22.7	45.0	2	10	343.5	-52	69	253			1
Isalo I	Middle to Late Triassic	16.7	47.1	4	14	350.5	-42	74	284			1
Isalo I to III	Middle Triassic to Middle Jurassic			8	32			74.2	277.1	77.7	6.3	1
Sakamena III	Early Triassic	23.5	44.7	1	4	339	-36.5	70	319			2
Sakamena II	Early Triassic	23.9	44.5	2	8	334	-56.5	64	276.5			2
Sakamena I	Late Permian	23.1	45.0	4	19	153	+48.5	65	294			3
Sakamena I to III	Late Permian to Early Triassic			7	31			65.8	291.6	96.4	6.1	
Sakoa III	Early Permian	23.8	44.8	2	17	158.5	+65.5	60	254.5			2
Sakoa I	Late Carboniferous	23.8	44.7	3	16	138	+67.5	48	264			3
Sakoa I to III	Late Carboniferous to Early Permian			5	33			52.8	260.7	93.9	7.9	

Note: References: 1, Embleton and McElhinny (1975); 2, Razafindrazaka and others (1976); 3, McElhinny and Embleton (1976).

III, about 1,400 m thick) in a conformable sequence. Paleomagnetic results have been obtained from three sites in the Glacial Series (McElhinny and Embleton, 1976) and two sites in the Lower Red Series (Razafindrazaka and others, 1976; Table 1). The Sakoa is followed unconformably by the Sakamena Group. In the southwest, the Upper Permian Lower Sakamena Beds (Sakamena I) are about 2,500 m thick and were sampled at four sites (McElhinny and Embleton, 1976). The Middle Sakamena, containing as much as 400 m of shales and limestones, including Early Triassic fish remains, yielded results at two sites, and the continental Upper Red Series (Sakamena III), about 600 m thick, produced data from a single site (Razafindrazaka and others, 1976; Table 1). The Isalo Group unconformably and transgressively overlying the Sakamena Group has been sampled at eight sites (Embleton and McElhinny, 1975). Red sandstones, shales, and mudstones were studied both in the southwest (Isalo II) and northwest (Isalo I and III) parts of the island.

All the paleomagnetic data were derived using thermal demagnetization techniques, sometimes with the addition of alternating field demagnetization. The data are summarized in Table 1, and an overall mean giving unit weight to sites has been calculated for each of the three groups. The overall means for both the Sakoa and Sakamena Groups have been greatly strengthened by combining the results obtained in the two laboratories.

COMPARISON WITH AFRICA AND SOUTH AMERICA

Paleomagnetic data from Africa covering the time span of the Karroo Supergroup are heavily biased in the age range Middle Triassic to Middle Jurassic, corresponding to the age of the Isalo Group. However, paleomagnetic data from South America are heavily biased in the time span late Carboniferous to Early Triassic, corresponding to the ages covered by the Sakoa and Sakamena Groups. Daly and Pozzi (1976) have noted that if the African and South American data are combined after reconstructing the Atlantic, there is a continuous distribution of poles covering the Permian and Triassic. To effect a comparison of the Madagascar results of Table 1 with those from Africa, we have used the observation of Daly and Pozzi and combined data from Africa and South America after rotating all South American poles to Africa, using the Smith and Hallam (1970) reconstruction (rotation about an Euler pole at 44°N, 329.4°E by an angle of 57°). K-Ar dates at 295 m.y. on

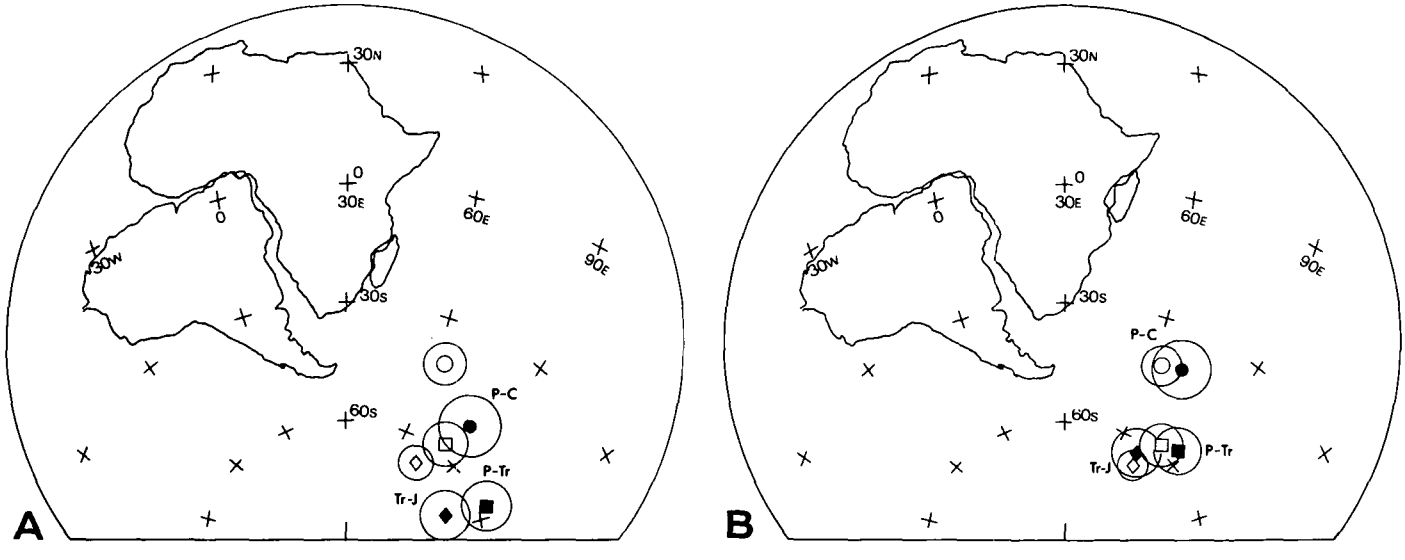


Figure 1. Paleomagnetic poles for three age groups from Madagascar (solid symbols) compared with combined data of Africa and South America (open symbols). Circles of 95 percent confidence are shown. A, Madagascar poles rotated to Flores (1970) reconstruction. B, Madagascar poles rotated to Smith and Hallam (1970) reconstruction.

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TABLE 2. COMPARISON OF PALEOMAGNETIC POLES FROM MADAGASCAR WITH COMBINED DATA OF AFRICA AND SOUTH AMERICA ACCORDING TO THREE HYPOTHESES FOR ITS PALEOPOSITION

Age	Pole positions: Africa - S. America (°S)	(°E)	Madagascar (°S)	(°E)	Angular differences (°): Present	Smith-Hallam	Flores
Triassic-Jurassic	66.0	71.8	74.2	97.1	11.8	3.5	15.1
Permian-Triassic	58.6	78.3	65.8	111.6	16.9	4.4	18.5
Permian-Carboniferous	40.8	62.7	52.8	80.7	17.2	5.6	16.8

the La Colina basalt (Thompson and Mitchell, 1972) now place the base of the La Colina formation (middle Paganzo) into the upper Carboniferous. For purposes of comparison with the Sakoa of Madagascar, we have combined results from the middle Paganzo of South America (Thompson, 1972) with Lower Permian data from Morocco (Daly and Pozzi, 1976) and Tanzania (Opdyke, 1964). This gives a combined Permian-Carboniferous pole for Africa and South America at lat 40.8°S, long 62.7°E ($N = 9$, $k = 108.0$, $\alpha_{95} = 4.9°$), if one uses the present-day coordinates of Africa after rotation of South America. There are only two results from the Lower Triassic of Africa (McElhinny and Brock, 1975) and none for the Upper Permian. However, when they are combined with mid-Permian to Early Triassic poles for South America (Valencio and Vilas, 1972; Veldkamp and others, 1971; Valencio and others, 1975), including results from the upper Paganzo (Thompson, 1972), we obtain a well-defined Permian-Triassic pole after rotation of South America at lat 58.6°S, long 78.3°E ($N = 10$, $k = 78.3$, $\alpha_{95} = 5.4°$). For the Middle Triassic to Middle Jurassic time span there are only four results from South America (Valencio and Vilas, 1972; McDonald and Opdyke, 1974; Vilas, 1974), but when these are combined with the 18 results now available from Africa (Daly and Pozzi, 1976; McElhinny and Brock, 1975; Dalrymple and others, 1975; Gidsgehaug and others, 1975), we get a combined Triassic-Jurassic pole at lat 66.0°S, long 71.8°E ($N = 22$, $k = 72.0$, $\alpha_{95} = 3.6°$).

The three overall mean poles from the Sakoa, Sakamena, and Isalo Groups (Table 1) may now be compared with the combined African and South American data outlined above. Constructing the position of Madagascar according to either Smith and Hallam (1970) or Flores (1970) gives the angular differences between corresponding pole pairs listed in Table 2 and illustrated in Figure 1. All the pole pairs are significantly different (average difference, 16.8°) on the Flores reconstruction (Fig. 1A), but they converge to differ by an average of only 4.5° on the Smith and Hallam reconstruction (Fig. 1B) when there are no significant differences. If Madagascar remains in its present position relative to Africa, the polar differences average 15.3°, which is hardly any improvement on the Flores position. These comparisons provide unequivocal support for the validity of the Smith and Hallam (1970) reconstruction originally proposed by du Toit (1937).

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